line and the neutral. Such a flexibility is not available in Delta connections.

In star connection, the neutral point can be earthed. Moreover, earthing of neutral permits to use protective devices to protect the system in case of ground faults.

5.6 DIGITAL ENERGY METER

Digital energy meters display the energy used on an LCD or LED display, and some can also transmit readings to remote places. In addition to measuring energy used, these meters can also record other parameters of the load and supply, such as instantaneous and maximum demands, voltage, power factor and reactive power etc. They can also support to record the amount of energy used during on-peak and off-peak hours.

The digital energy meter has a power supply, metering circuit, a processing and communication circuit (microprocessor/microcontroller) and other add on modules such as RTC (Real Time Clock), LCD (Liquid Crystal Display), communication ports, modules and so on.

In the metering circuit, the voltage and current inputs through current transformer and potential transformer has a voltage reference followed by an ADC (Analog to Digital Convertor) section, to convert the analog inputs into digital forms. These inputs are then processed using a digital signal processor, to calculate various metering parameters.

The processing and communication section has the responsibility of calculating various desired quantites, from the digital values generated by the metering section. This has the responsibility of communicating and interfacings with other 'add on modules' connected as slaves to it.

RTC (Real Time Clock) and other add-on modules are attached as slaves to the processing and communication sections

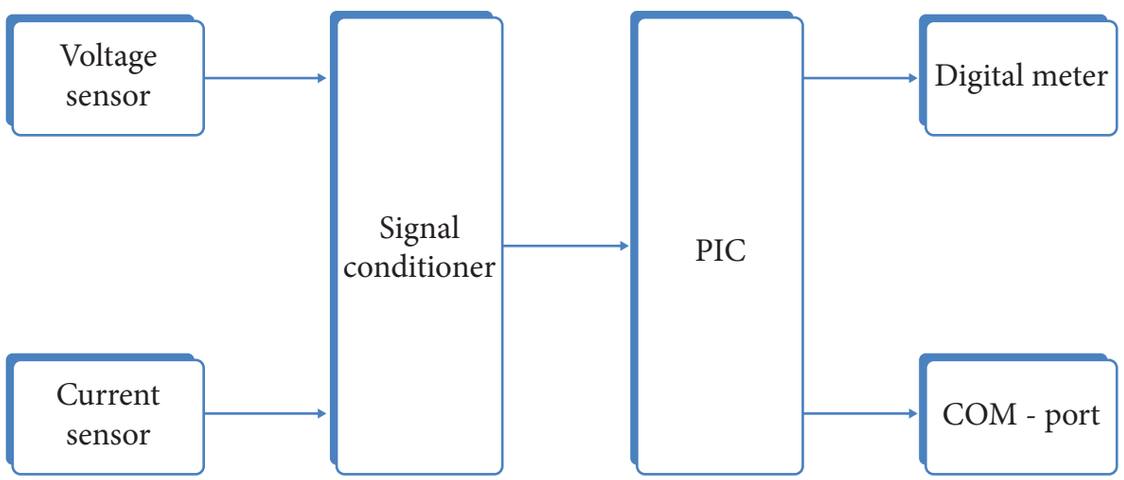


Fig 5.16 *Block Diagram of Digital Energy Meter*

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